



Expected Goals (xG) Modelling in Football

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1. Abstract

Expected Goals (xG) estimates the probability that a football shot results in a goal. This project builds a calibrated xG model on **StatsBomb open data** as a mathematical modelling exercise: shot geometry is derived by trigonometry, the baseline model is framed as maximum likelihood estimation with interpretable coefficients, and all models — logistic, Random Forest, XGBoost — are evaluated on discrimination and calibration against StatsBomb's own xG. Shot-level xG is then aggregated to team-season totals, checking the model's predictions against actual 2015/16 league standings.

2. Data Scope

StatsBomb's open data [StatsBomb 2024a] has only **one fully covered season** for any major domestic league — every other season is a partial, player-specific subset (heavily skewed toward Barcelona/Messi) that would bias a distance/angle model. This project combines the four leagues with a complete **2015/16** season:

Leagues: La Liga, Premier League, Serie A, Ligue 1

Matches: 1,517

Shots (penalties excluded): 37,488

Goals: 3,569 (9.5% conversion rate)

3. Pipeline

Extraction: fixtures and shot events pulled via statsbombpy [StatsBomb 2024b].

Geometry: shot distance and angle derived by trigonometry.

Baseline model: logistic regression on distance + angle, fit by MLE.

Enhanced model: add body part, technique, play pattern, pressure, first-time.

Tree-based models: Random Forest and XGBoost, same feature set.

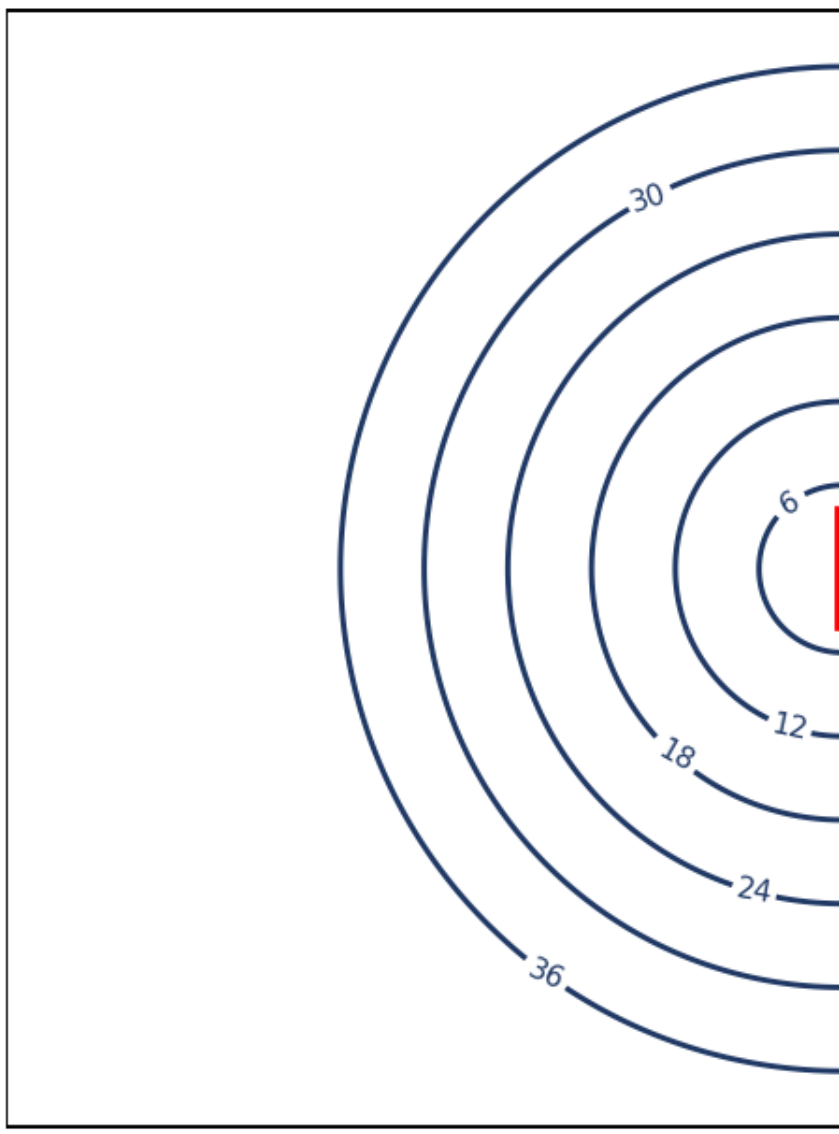
Evaluation: AUC-ROC, Brier score, log-loss, Expected Calibration Error.

4. Shot Geometry

Distance and angle to goal are derived from StatsBomb's (x, y) shot-location coordinates on a 120×80 pitch, relative to the goalposts $P_1 = (120, 36)$ and $P_2 = (120, 44)$:

$$\text{distance} = \sqrt{(120 - x)^2 + (40 - y)^2}, \quad \theta = \left| \arctan\left(\frac{44 - y}{120 - x}\right) - \arctan\left(\frac{36 - y}{120 - x}\right) \right|$$

Iso-distance curves (units to goal centre)



Iso-angle curves (goal width seen, degrees)

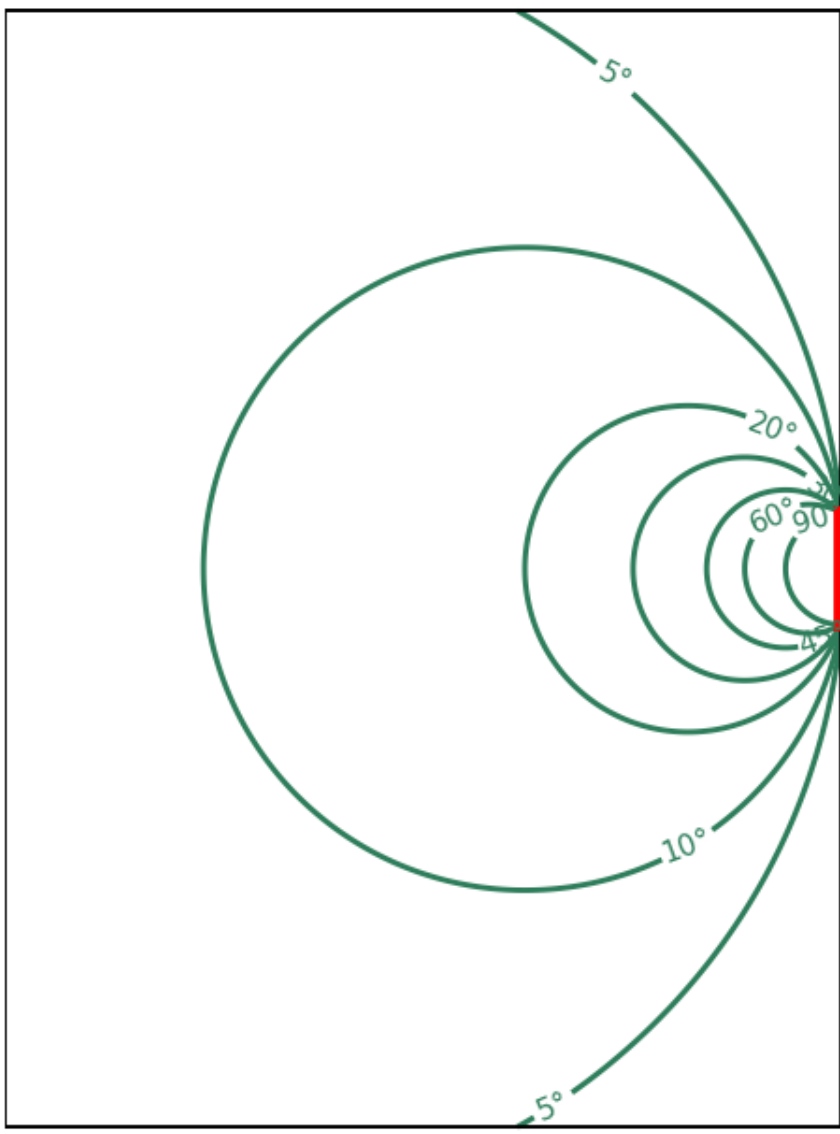


Figure 1. Iso-probability curves in pitch space as a function of shot geometry.

5. Feature Importance

Random Forest importances confirm that geometry dominates the model: distance and angle jointly account for $\approx 79\%$ of total importance, with shot context (header, first-time, pressure) making up the remainder.

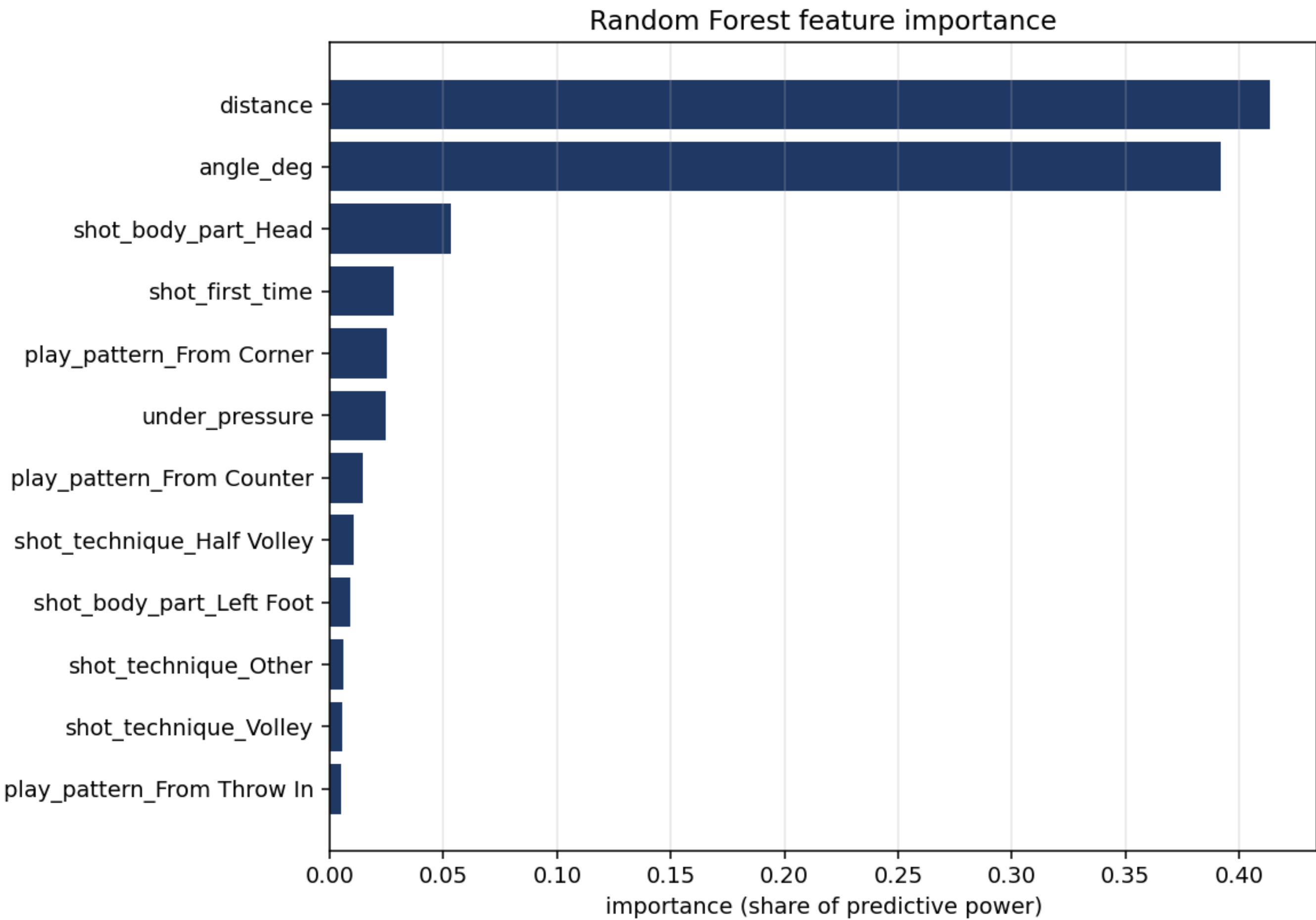


Figure 2. Random Forest feature importances.

6. Model Progression

Baseline logistic regression — distance + angle only; interpretable by construction.

■ Coefficients reported as log-odds and odds ratios, each with a p -value.

■ Every additional metre from goal multiplies scoring odds by ≈ 0.93 .

Enhanced logistic regression — adds categorical shot context via one-hot encoding.

Random Forest & XGBoost — trade interpretability for flexibility; compared on identical features and evaluation criteria.

7. Baseline Model: Maximum Likelihood Formulation

The baseline model is fit as a logistic regression, estimating β by maximising the likelihood of the observed goal/no-goal outcomes:

$$P(\text{goal} \mid \text{dist}, \theta) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \text{dist} + \beta_2 \theta)}}$$

Feature	Coef. (log-odds)	Odds ratio	p -value
Distance (m)	-0.0733	0.929	< 0.001
Angle (deg.)	+0.0229	1.023	< 0.001

Table 1. Baseline logistic regression coefficients.

8. Results: Model Comparison

Model	AUC	Brier	LogLoss	ECE
Baseline logit	0.746	0.0795	0.2803	0.0162
Enhanced logit	0.777	0.0772	0.2702	0.0138
Random Forest	0.779	0.0769	0.2690	0.0126
XGBoost	0.779	0.0769	0.2688	0.0114
StatsBomb xG	0.816	0.0714	0.2510	0.0116

Table 2. Model comparison on the held-out test set.

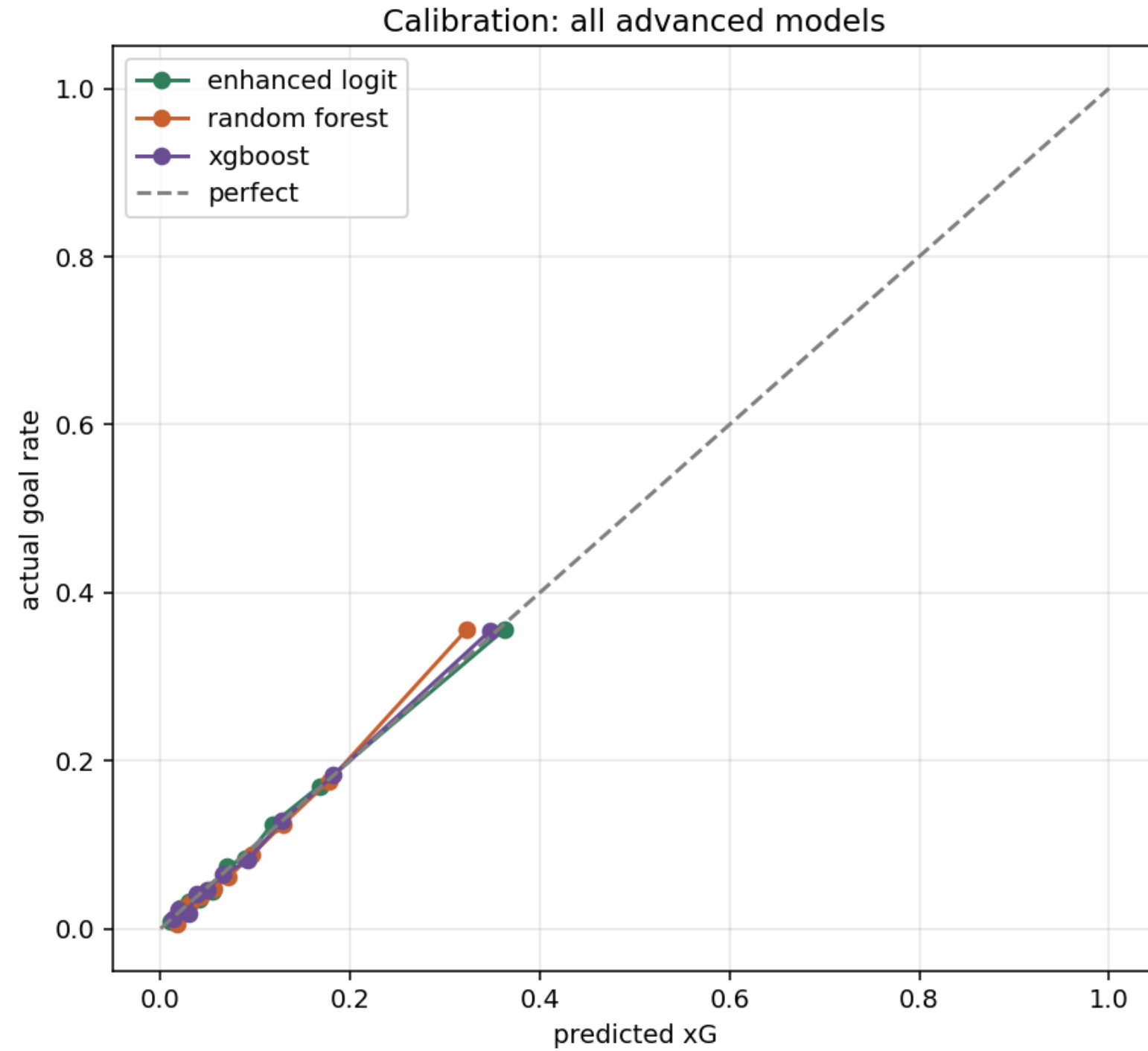


Figure 3. Calibration curves for all models against the diagonal (perfect calibration).

9. Applied Outputs

Player finishing (goals - xG) surfaces over-performing finishers relative to shot quality: L. Suárez (+14.0), Z. Ibrahimović (+13.8), and G. Bale (+10.8) top the 2015/16 sample across all four leagues.

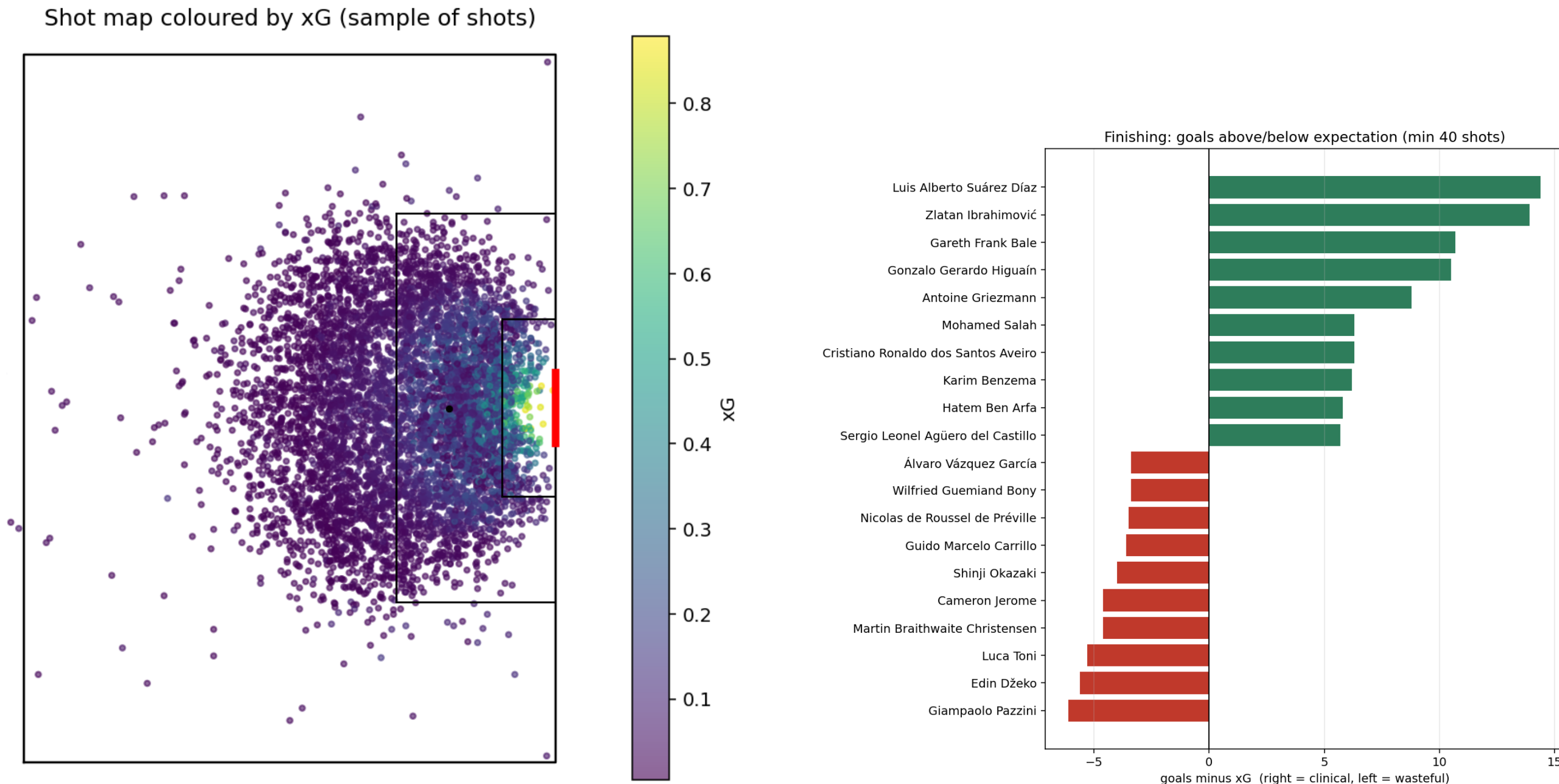


Figure 4. *Left:* shot map coloured by predicted xG. *Right:* player finishing table (goals - xG).

10. Team-Level Expected Points

Aggregating shot-level xG to team-season totals converts the model into a checkable, real-world claim: how many points a team's chance quality alone would predict, versus what they actually earned across the 2015/16 season.

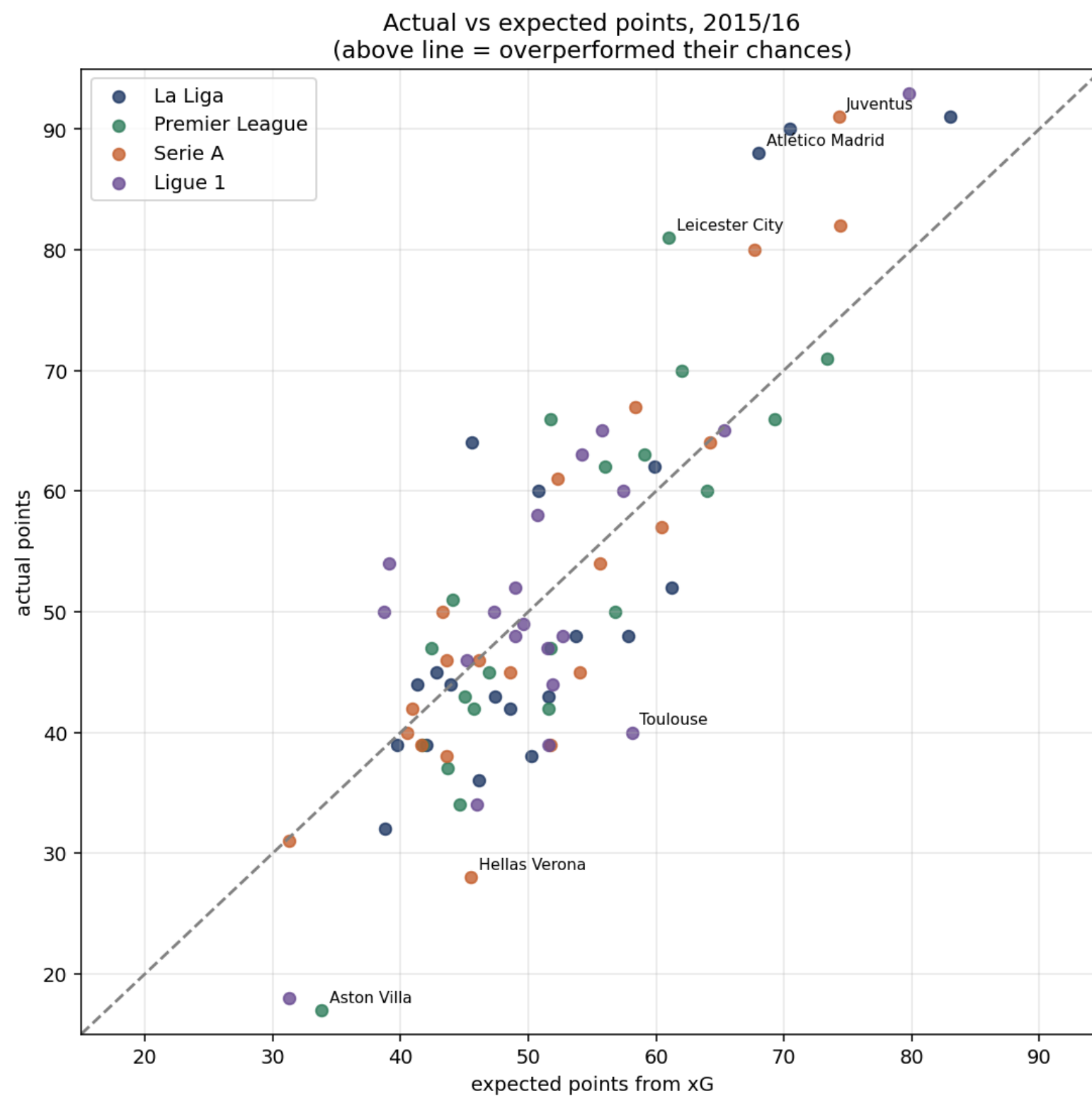


Figure 5. Actual vs. expected points, 2015/16. Leicester City's title-winning overperformance (+20.4 points) and Aston Villa's relegation-season underperformance are both visible relative to the diagonal (perfect agreement).

11. References

StatsBomb (2024). *StatsBomb Open Data*. <https://github.com/statsbomb/open-data>
StatsBomb (2024). *statsbombpy: A Python Client for StatsBomb Data*.
<https://github.com/statsbomb/statsbombpy>

12. Project Repository



Scan to view the full code, data, and report: github.com/ACM40960/projects-rishveed-vishesh